

Discovery, Prediction, and Explanation in Active Learning for Protein–Lipid Binding Affinity

Editors: Andy Song, Bo Han and Sarah Erfani

Abstract

Active learning for molecular affinity can prioritize strong binders without necessarily improving prediction over the full candidate distribution. We examine this distinction in a retrospective protein–lipid benchmark assembled from published measurements. Two pooled BioDolphin endpoints contain 864 and 549 deduplicated protein–lipid pairs, while four small target panels contain 11–16 pairs each. We evaluate five lipid representations, including three frozen molecular transformers, five Gaussian-process kernels and 13 query policies, yielding 5,850 campaigns across three seeds. Uncertainty-only acquisition gives the lowest descriptive mean MAE on pooled dissociation and inhibition affinities (0.641 and 0.429 pK), whereas a Thompson configuration on dissociation affinities recovers 96.7% of the pool’s top-decile pairs with MAE 0.705. Prediction and discovery rankings therefore disagree. We complement final-model grouped SHAP with 14,754 ungrouped ECFP explanations across 396 cycle snapshots, including affinity-prioritized fragment visualization and protocol stability analysis. Exact affine-model attributions agree with independent KernelExplainer checks to below 10^{-10} pK. These checks establish computational fidelity, not mechanistic validity. Random-pair splits, heterogeneous assays, small target panels, and post-hoc configuration selection limit generalization claims. The study provides an auditable domain adaptation and exposes the distinction between acquisition success, predictive quality, and chemical interpretation. A companion dashboard exposes the complete grid and local measured-data campaigns.

Keywords: active learning; protein–lipid affinity; Gaussian processes; Thompson sampling; SHAP; retrospective evaluation

1. Introduction

Protein–lipid interactions present a useful setting for evaluating molecular active learning: experiments are limited, the candidate space is chemically heterogeneous, and collecting strong binders is not the same objective as predicting every affinity accurately. A policy that repeatedly selects high-scoring pairs may enrich its measured set while neglecting regions of the distribution needed for accurate global prediction. This distinction should be made explicit when evaluating an acquisition policy.

Published protein–lipid resources enable retrospective evaluation, but their size can be misleading if structural contacts are treated as independent quantitative measurements. BioDolphin supplies a broad interaction resource (Yang et al., 2024); only its curated affinity-bearing subset is used here. Separate TRAAK measurements provide much smaller, target-specific panels (Schrecke et al., 2021). These sources offer complementary evaluation regimes, not six independent large-scale biological studies.

We adapt the ligand-affinity framework of Srivastava et al. (2026) to pairwise protein–lipid prediction. This is an empirical transfer and audit rather than a new acquisition algorithm. Our contributions are: (i) a matched representation–kernel–policy grid on measured

protein–lipid affinities; (ii) separate evaluation of held-out prediction and acquisition-pool enrichment; (iii) cycle-resolved, chemically annotated explanations with explicit numerical and interpretive checks; and (iv) an interactive results portal that separates complete benchmark coverage from selected SHAP analyses. Negative findings on tiny panels and disagreement between predictive and discovery objectives are retained, rather than obscured by reporting only the most favorable scores.

2. Related Work and Scope

Gaussian-process molecular active learning combines a surrogate mean with uncertainty-aware querying. The reference framework compares molecular encodings, covariance functions and scheduled exploration/exploitation (Srivastava et al., 2026). We retain its grid structure and add joint-posterior Thompson sampling (Russo et al., 2018): a correlated function realization is drawn over the remaining pool, and the highest sampled values form the next batch. The method does not replace that joint draw by independent uncertainty noise.

SHAP provides additive attributions relative to a specified background game (Lundberg and Lee, 2017). In molecular applications, assigning importance to a fingerprint bit does not uniquely identify a chemical fragment because hash collisions and repeated environments are possible. Furthermore, feature replacement can leave the manifold of valid molecules. We separate numerical attribution checks, structural annotations, and biological claims. No prospective experiment, protein-contact analysis or validated binding pose is provided. Neither learned protein embeddings nor a newly trained deep kernel is evaluated. ChemBERTa-MTR, ChemBERTa-MLM and MoLFormer are frozen lipid encoders; their use is not a claim of a novel transformer.

3. Data and Evaluation Design

3.1. Measured datasets

Table 1 lists the benchmark subsets. Positive quantitative affinities are converted to $pK = -\log_{10}(K/M)$, so higher values indicate stronger affinity. The pooled K_d and K_i endpoints remain separate. Deduplication is by protein sequence and lipid SMILES; the curated median is an aggregation of source records, not a newly measured affinity. Censored annotations are excluded. BioDolphin’s broad lipid definition is not restricted to membrane phospholipids. Its imported assay annotations may originate in other affinity resources; this benchmark does not claim that all measurements were generated by BioDolphin itself.

TRAAK panels use the mean first sequential binding constant, K_{d1} , from the published triplicates. Replicates and subsequent binding events are not counted as additional training pairs. Two transthyretin (TTR) sequence-defined subsets are drawn from BioDolphin and overlap its pooled K_d panel. The two publications are the independent data sources; endpoints, constructs and subsets are not independent replications.

Table 1: Curated benchmark subsets and acquisition budgets. TTR-A and TTR-B denote the sequence-defined subsets with identifiers 889201f64e9f and e1eca46439c9, respectively. Small panels are exploratory.

Panel	Pairs	Test pairs	Initial labels	Final labels
BioDolphin K_d	864	173	60	360
BioDolphin K_i	549	110	60	360
TRAAK-a K_{d1}	16	3	2	8
TRAAK-b K_{d1}	16	3	2	8
TTR-A K_d	11	2	2	8
TTR-B K_d	11	2	2	8

3.2. Splits, budgets, and metrics

Seeds 7, 19 and 42 define matched random-pair partitions. Approximately 20% of rows are frozen as test data; test labels never guide querying or model fitting. Each pooled campaign begins with 60 labels and acquires ten batches of 30. This measures 360 of 691 eligible K_d pairs and 360 of 439 eligible K_i pairs. The latter budget covers about 82% of its pool, making high final recall less demanding than early discovery. Small panels start with two labels and distribute six further labels over ten phase slots as (0, 1, 0, 1, 1, 0, 1, 0, 1, 1). Zero-budget phases do not refit the model. Their schedules preserve phase positions but do not reproduce the large-panel measurement budget.

We report held-out MAE in pK units and R^2 , separately from top-decile recall in the acquisition pool. The hit set contains the top $\lceil 0.1N_{\text{pool}} \rceil$ affinities, including ties at the threshold; initial labels count toward recall. Recall AUC is the trapezoidal area under recall versus number of labels, divided by the acquired-label range. It captures discovery timing rather than only the endpoint. The experiments also record top-2% and top-5% recall, RMSE, simple regret and 90% predictive-interval coverage.

Configurations share test partitions and initial labels within a seed. Consequently, configurations are not independent replicates. Reported means and standard deviations refer to the three seeds. Selecting a configuration by its observed test MAE or recall AUC makes it a descriptive winner, not an unbiased estimate of a validated model-selection procedure. No significance claim is made from three seeds or overlapping panels.

4. Surrogates and Acquisition

4.1. Representations and pairwise covariance

Lipid inputs are radius-4, 4096-bit Morgan fingerprints (ECFP; [Rogers and Hahn, 2010](#)), RDKit’s 167-output-bit MACCS representation, fixed 384-dimensional ChemBERTa MTR and MLM CLS embeddings (77M checkpoints) ([DeepChem, n.d.](#)), and 768-dimensional masked-mean MoLFormer-XL embeddings ([Ross et al., 2022](#)). Radius four follows the reference implementation and corresponds to diameter-eight terminology; we do not silently relabel it as radius-two ECFP. All 549 distinct lipid strings are encoded without trunca-

tion: maxima are 156 ChemBERTa and 154 MolFormer tokens. The latter checkpoint is pretrained on 10% of both ZINC and PubChem; canonical non-isomeric SMILES match its stereochemistry-removing preprocessing. This loss of stereochemical information is a representation limitation. The encoders are not affinity-fine-tuned. Revisions and feature hashes are pinned; all pretrained base-model tensors load strictly. Protein inputs contain 20 amino-acid composition values, four composition groups, log sequence length and a charge descriptor.

For a pair $z = (p, l)$ we use

$$k(z, z') = s k_L(l, l') k_P(p, p'), \quad (1)$$

where k_L is Tanimoto, linear, RBF, rational quadratic, or Matérn-3/2. The protein factor is Matérn-3/2 and becomes one on a constant-protein panel. Protein standardization and distance scaling use initial inputs only and remain fixed through the campaign. The dot-product Tanimoto extension is also evaluated on dense transformer vectors; it is not a binary chemical-substructure interpretation of those coordinates.

An exact GP (Rasmussen and Williams, 2006) learns a constant mean, signal variance and noise variance. Applicable length scales and the rational-quadratic shape are also learned by marginal likelihood optimization. Target normalization uses acquired labels only. Optimization uses analytic-gradient L-BFGS-B, warm starts and at most 60 iterations per fit. This differs from the reference authors’ Adam routine; the protein covariance factor is also a domain adaptation. Posterior means and variances, hyperparameters, optimization status and row indices are saved at every fitted round.

4.2. Query policies

Write E for ranking by observation predictive standard deviation, X for ranking by predicted mean, and B for ranking by $0.5\mu + 0.5\sigma$. The schedules are balanced (B^{10}), alternate ($(EX)^5$), sandwich ($E^2X^6E^2$), explore-heavy (E^7X^3), exploit-heavy (X^7E^3), and gradual ($E^3B^4X^3$), together with uniform random selection. Thompson sampling draws

$$\tilde{\mathbf{f}} \sim \mathcal{N}(\boldsymbol{\mu}_{\mathcal{U}}, \boldsymbol{\Sigma}_{\mathcal{U}}) \quad (2)$$

over the unqueried pool $\mathcal{U}$, using latent rather than observation covariance, and selects the top batch. A small adaptive Cholesky jitter stabilizes the joint draw.

The expanded design adds pure exploitation (X^{10}), pure uncertainty (E^{10}), expected improvement (EI), probability of improvement (PI), and a Thompson–diversity heuristic. For acquired-label incumbent y^* , $d = \mu - y^* - \xi$, $z = d/\sigma_f$, and $\xi = 0.01$ pK, EI is $d\Phi(z) + \sigma_f\varphi(z)$ and PI is $\Phi(z)$; σ_f is latent posterior SD and zero-variance limits are handled explicitly. Batches select the largest marginal scores, not joint qEI or noise-integrated EI. The diversity heuristic standardizes one joint Thompson draw and greedily adds $0.25[1 - \exp(-\min_{j \in S} d_L(i, j)^2/2)]$ to each remaining score, where S is the within-batch selected set; the first selection has zero bonus. The distance is initial-input-scaled lipid distance. Constants were fixed before the expanded runs, not tuned on held-out outcomes. This is a heuristic, not a new acquisition algorithm or a state-of-the-art claim.

The full design comprises $6 \times 5 \times 5 \times 13 \times 3 = 5850$ campaigns, or 1,950 three-seed configurations. It reuses 2,160 hash-verified original campaigns and adds 3,690 runs, preserving

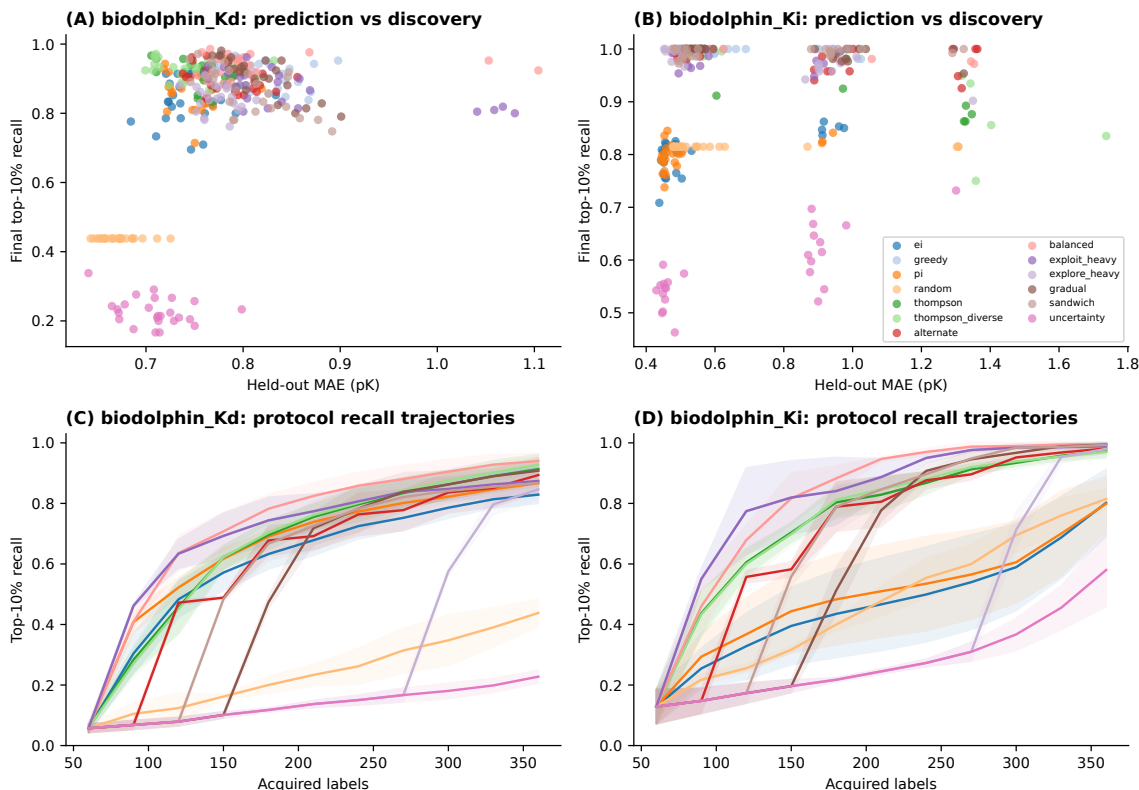


Figure 1: Four-panel performance overview. (A,B) Each point is a three-seed configuration mean, showing that low held-out MAE and high acquisition recall are different objectives. (C,D) Within each seed, recall is averaged over the 25 representation-kernel configurations; lines and shading then show the three-seed mean and standard deviation. Shading is descriptive variability, not a confidence interval.

their splits, budgets and GP engine. All supported combinations are evaluated; this finite suite is not exhaustive of the acquisition-function or kernel literature.

5. Explainability Protocol

5.1. Final-model grouped analysis

For each panel, the original 2,160-run grid’s descriptive prediction winner, discovery winner and lowest-MAE Thompson configuration were explained across three seeds: 54 saved final models. This explanation cohort is retained, not reselected using the expanded grid. No SHAP coverage is claimed for all 5,850 runs or for the new MLM/MoLFormer models. Up to eight test pairs are explained using up to eight acquired-training background pairs. The exact two-player game separates protein and lipid contributions. A finer permutation-SHAP game uses 20 training-input-variance-selected fingerprint bits, a joint remainder-bit group and a joint protein group. ChemBERTa uses 12 embedding blocks instead. Two

independent runs of eight antithetic permutations provide a sampling-sensitivity diagnostic, not an attribution confidence interval. Fine-group values and two-player values are different games and are not added together. Embedding blocks are not named chemical motifs.

5.2. Cycle-resolved individual-bit explanations

To follow the temporal chemical analysis of [Srivastava et al. \(2026\)](#), we compare matched ECFP-linear explore-heavy and exploit-heavy campaigns on all six panels and all three seeds. We explain the initial model and ten subsequent cycle snapshots: 36 campaigns and 396 snapshots. Each cycle draws up to 100 unqueried pairs, excluding frozen test rows, and up to 50 acquired-training lipids as background. Small panels use all available observations. All 4096 bits receive individual attributions before the top ten are selected by mean absolute SHAP.

For pooled data, the query protein is fixed while background lipids are substituted. The linear lipid kernel then yields an affine posterior mean $f_p(x) = a + \mathbf{w}(p)^\top x$. For the empirical background mean $\bar{x}$, the exact marginal-game Shapley values are

$$\phi_j(x; p) = w_j(p)(x_j - \bar{x}_j), \quad \phi_0(p) = a + \mathbf{w}(p)^\top \bar{x}. \quad (3)$$

This follows because replacing any subset of coordinates changes the affine prediction by the sum of their separate effects; every ordering therefore gives the same marginal contribution. The formula explains lipid variation conditional on the query protein, not protein main effects. It cannot be applied to the non-linear lipid kernels.

Equation 3 avoids a sampled Kernel SHAP computation for every pair. One final-cycle pair per campaign is independently checked with `KernelExplainer`, using $2M + 128$ coalition samples for M varying coordinates and no L1 feature selection. Replacing the background by its mean preserves every coalition value exactly for the affine predictor. This is a disclosed computational shortcut, not a claim to reproduce the reference implementation’s unspecified regularization settings.

5.3. Fragments and stability

At each cycle, importance is $N^{-1} \sum_i |\phi_j(x_i; p_i)|$. Evolution plots follow final top-ranked bits across all cycles, while per-cycle top-ten lists are also saved. For a selected bit, measured bit-containing pairs are sorted by descending affinity and an RDKit environment from the highest-affinity molecule is highlighted. All environments for that bit in the representative molecule are retained. We rank fragments by pair prevalence multiplied by mean absolute SHAP; the algebra of this combined score is an explicit implementation choice.

Affinity distributions for present and absent bits use all measured rows retrospectively, including held-out rows. These labels do not enter SHAP, top-bit selection, querying or fitting, but they do enter the post-hoc fragment display and cannot serve as independent validation. Protocol and consecutive-cycle stability are measured using top-ten Jaccard overlap and Spearman correlation over bits active in either importance vector. Changing the sampled background or unqueried population can change importance even with an unchanged GP.

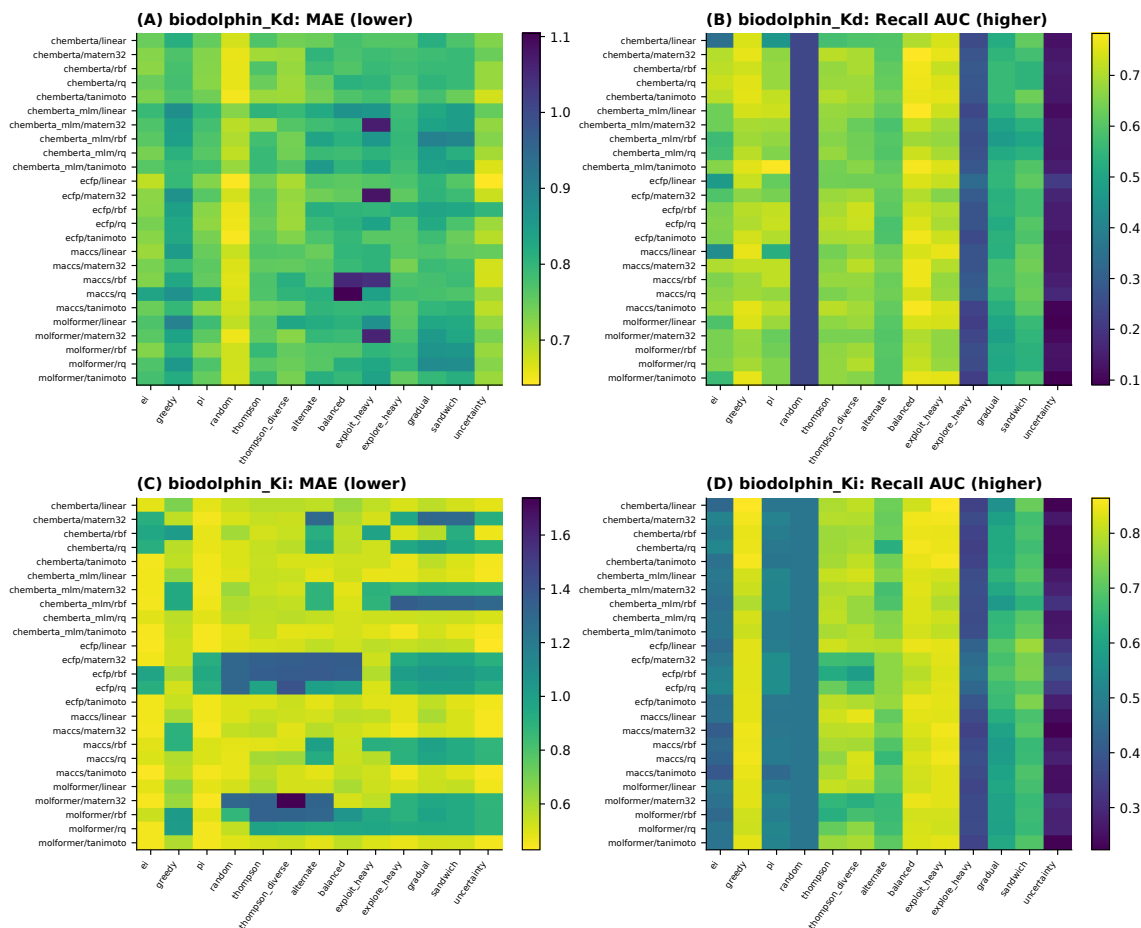


Figure 2: Complete pooled-data representation-kernel grids in four panels. (A,C) Final test MAE. (B,D) Label-normalized top-decile recall AUC. Rows are representation/kernel combinations and columns are query policies. All values are three-seed means; pooled K_d and K_i retain their separate endpoint definitions.

6. Results

6.1. Prediction and discovery select different configurations

Table 2 and Figures 1–2 show the central trade-off. For pooled K_d , ECFP-linear with uncertainty-only selection achieves the lowest observed mean MAE, 0.641 ± 0.041 , and mean $R^2 = 0.665$. Its final recall is 0.338. The original grid’s random-acquisition winner remains close at 0.643 ± 0.055 ; this small post-hoc difference is not evidence of statistical superiority. The lowest-MAE Thompson configuration, ChemBERTa-MTR-Matérn, has MAE 0.705 ± 0.046 and recall 0.967. Thus high-affinity discovery does not require the lowest global test error, and maximizing discovery need not optimize that error.

The highest recall-AUC K_d configuration combines balanced UCB and ChemBERTa-MLM-linear, yielding AUC 0.783 and final recall 0.976. This is distinct from choosing

Table 2: Descriptive selected pooled configurations (three seeds). P: lowest mean MAE; D: highest mean recall AUC; T: lowest mean MAE among Thompson models. MTR/MLM denote ChemBERTa variants; M32 is Matérn-3/2. Selection is post-hoc on these results, not nested validation.

Endpoint/role	Representation/kernel	Policy	MAE	R^2	Recall	AUC
K_d /P	ECFP/linear	Uncertainty	.641	.665	.338	.210
K_d /D	MLM/linear	Balanced	.868	.467	.976	.783
K_d /T	MTR/M32	Thompson	.705	.624	.967	.711
K_i /P	ECFP/linear	Uncertainty	.429	.846	.543	.319
K_i /D	MTR/linear	Exploit-heavy	.554	.787	1.000	.864
K_i /D	MTR/linear	Greedy	.689	.666	1.000	.864
K_i /T	MACCS/RBF	Thompson	.490	.836	1.000	.784

the largest endpoint recall. On K_i , the lowest-MAE configuration is ECFP-linear with uncertainty-only querying, giving MAE 0.429 ± 0.044 and mean $R^2 = 0.846$, compared with the previous lowest MAE of 0.455. Its recall AUC is 0.319 and final recall is 0.543. ChemBERTa-MTR-linear with greedy or exploit-heavy querying ties for highest AUC (0.864), with final recall 1.000. We retain both tied winners in the supplementary tables; exploit-heavy has lower MAE (0.554 versus 0.689). High endpoint recall alone can also conceal the difference in when high-affinity pairs were acquired.

These are comparisons at the same 360-label pooled budget. An earlier fixed-kernel benchmark used only 64 labels; its plots are retained in the supplementary atlas but are not evidence of budget-matched improvement. No claim of superiority over external state-of-the-art methods follows from the present grid.

6.2. Tiny target panels do not establish reliable prediction

Even the lowest-MAE selected configurations have negative mean R^2 on every small panel: -0.217 and -0.265 on TRAAK-a and TRAAK-b, and -2.601 and -1.138 on TTR-A and TTR-B. These are the lowest-MAE selections, not selections by maximum R^2 . Their test sets contain only two or three pairs. A small absolute MAE on a narrow affinity range does not establish useful ranking or extrapolation. Some final-model lipid attributions are nearly zero, consistent with weak discrimination on the explained examples. Protein attribution is zero on a fixed-protein panel by construction, not evidence that the protein is biologically irrelevant. Small-panel high recall is also easy to obtain when much of the acquisition pool has been measured.

6.3. Frozen transformers do not uniformly improve affinity learning

Figure 4 fixes Matérn-3/2 and Thompson sampling to compare all five encodings without changing policy or kernel. On K_d , mean MAEs are 0.759 (ECFP), 0.753 (MACCS), 0.705 (MTR), 0.711 (MLM), and 0.753 (MoLFormer). On K_i they are 1.329, 0.566, 0.542, 0.540, and 1.328, respectively. Thus the additional encoders do not monotonically improve prediction; the small MLM-MTR difference on K_i is descriptive. Representation rankings

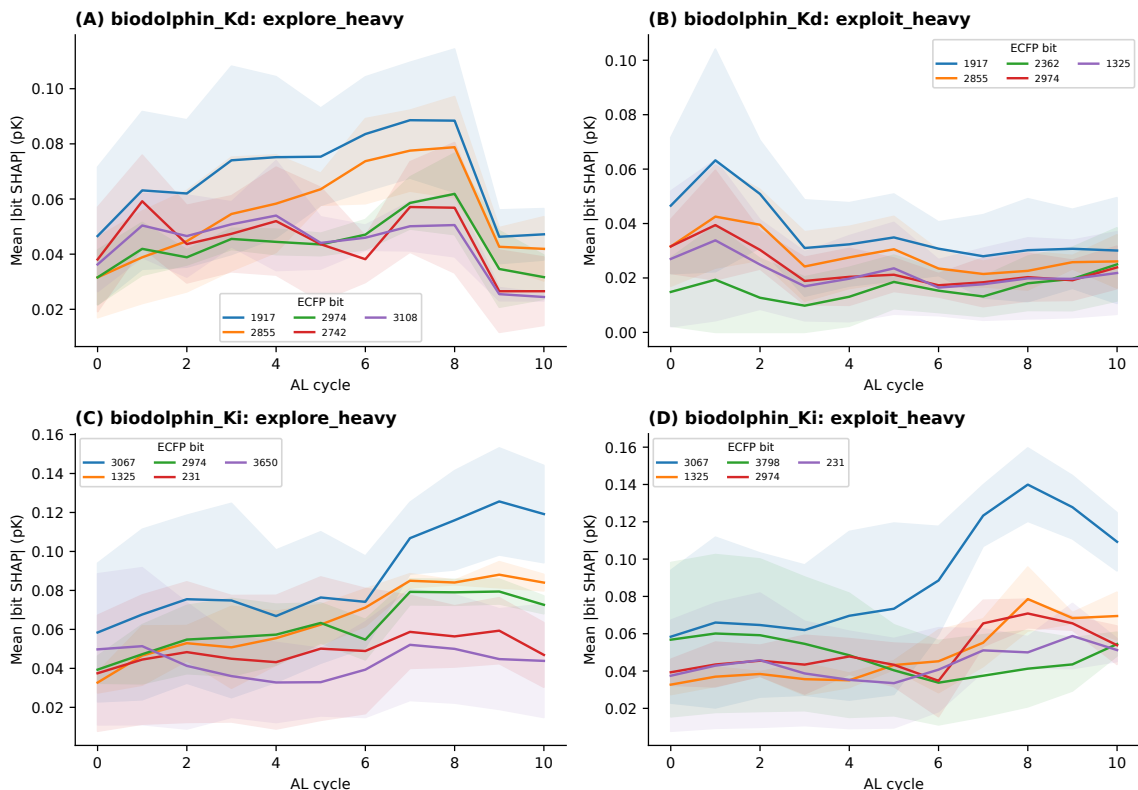


Figure 3: Four temporal explanation panels for matched ECFP-linear campaigns. The five bits with largest final importance averaged across seeds are shown in each panel. Lines are seed means; bands are seed standard deviations. Explore-heavy uses E^7X^3 and exploit-heavy uses X^7E^3 . Importance reflects model, query population and background together; it need not increase monotonically with measurement count.

depend on kernel and policy: the overall predictive winners on both pooled endpoints remain ECFP-linear with uncertainty-only acquisition. Frozen molecular pretraining, assay heterogeneity and coarse protein features are plausible contributors, not experimentally isolated causes.

6.4. Attribution fidelity and chemical interpretation

The grouped analysis explains 234 test-pair/model instances across 54 models. Rebuilt posterior means match saved predictions to at most 2.72×10^{-13} pK; additive reconstruction error is at most 1.30×10^{-13} pK. The largest model-level mean absolute difference between the two permutation-SHAP repetitions is 0.0262 pK. This checks sampling sensitivity at the chosen background, not robustness to alternative backgrounds or to a different model.

For selected Thompson models, mean absolute two-player contributions averaged across seeds are 0.802 pK for lipid and 0.835 pK for protein on K_d , and 1.012 versus 0.899 pK on K_i .

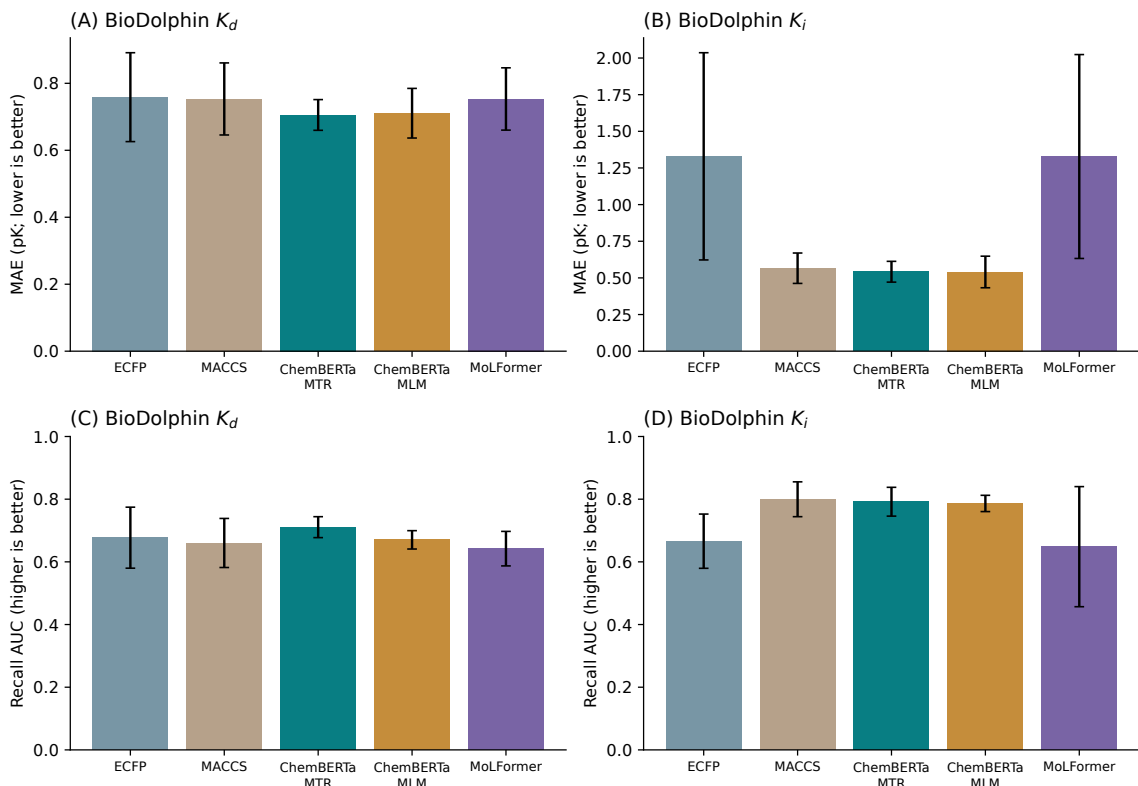


Figure 4: Matched representation comparison in four panels. All encoders use Matérn-3/2, joint Thompson sampling, identical splits and the same label budget. (A,B) Final held-out MAE. (C,D) Label-normalized discovery AUC. Bars and whiskers are three-seed means and sample SD, not confidence intervals. MTR and MLM denote the two frozen ChemBERTa checkpoints.

They are not percentages of predictive accuracy and should not be compared as causal effect sizes. The detailed temporal analysis yields 14,754 pair/model explanations. Its maximum additive reconstruction error is 9.86×10^{-14} pK; the 36 independent KernelExplainer checks disagree by at most 9.23×10^{-11} pK. Exact agreement is expected for the affine game and is not evidence of a correct biochemical model.

Temporal rankings vary (Figure 3). Final top-ten Jaccard overlap between explore-heavy and exploit-heavy averages 0.545 on both pooled endpoints. Active-bit rank correlations average 0.642 for K_d and 0.597 for K_i . These moderate agreements do not support protocol-independent motif rankings. The supplementary highlighted fragments provide concrete annotations of bit environments. Their affinity-prioritized selection deliberately favors potent examples, so the resulting visual association is not an independent SAR test. Figure 5 gives four representative chemical panels; different important environments may be drawn from the same potent molecule.

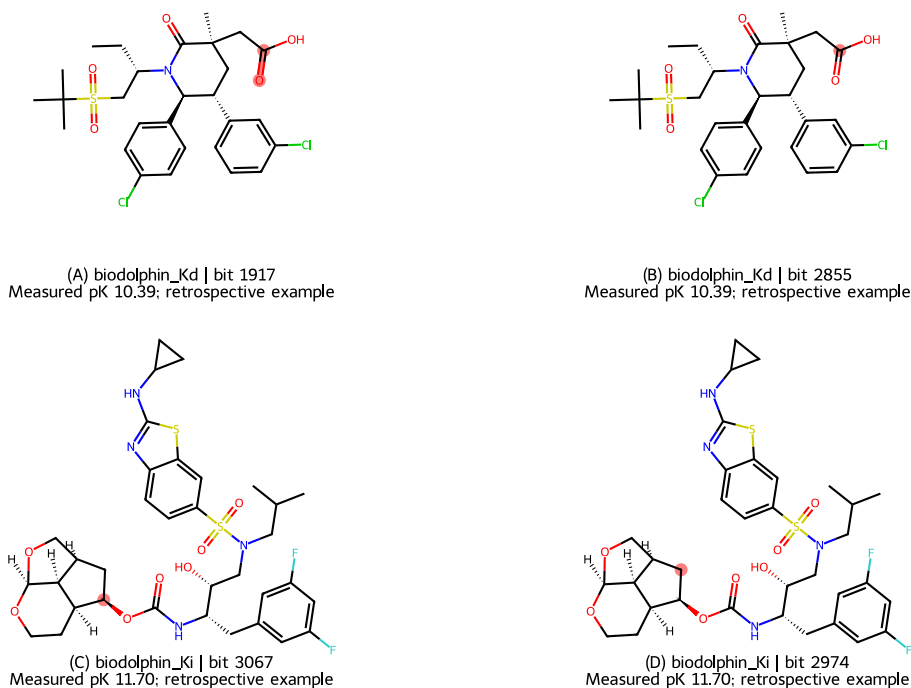


Figure 5: Four affinity-prioritized molecular examples from seed-7 ECFP-linear explore-heavy campaigns: two highest combined-score bits for pooled K_d (A,B) and K_i (C,D). Highlighted atoms mark one recorded ECFP environment, not a binding site. Whole molecules are shown to preserve chemical context. Examples use retrospective measured labels and may repeat the same molecule for different bits; they are not four independent hits or evidence of causal interactions. Full collision records appear in the atlas and source outputs.

6.5. Execution and reproducibility checks

All 5,850 expected runs are present. Split separation, matched initialization, phase schedules and label budgets pass the recorded audit. Of 48,750 fits, 48,724 meet the optimizer convergence criterion; 26 do not. These are explicitly logged, not silently described as converged. On the 54 selected explanation campaigns, reconstructing saved posteriors and replaying original random-number streams reproduces selected indices in all 396 nonempty acquisition rounds. Final-prediction SHAP is distinct from this acquisition replay: it does not explain a particular Thompson function draw. Independent tests cover covariance calculations, analytic gradients, saved-prediction agreement and masked-feature predictions. Additional checks exercise all 325 representation-kernel-policy combinations on measured TRAAK data, transformer determinism and padding invariance, and actual browser upload campaigns. These are computational checks, not evidence of model generalization.

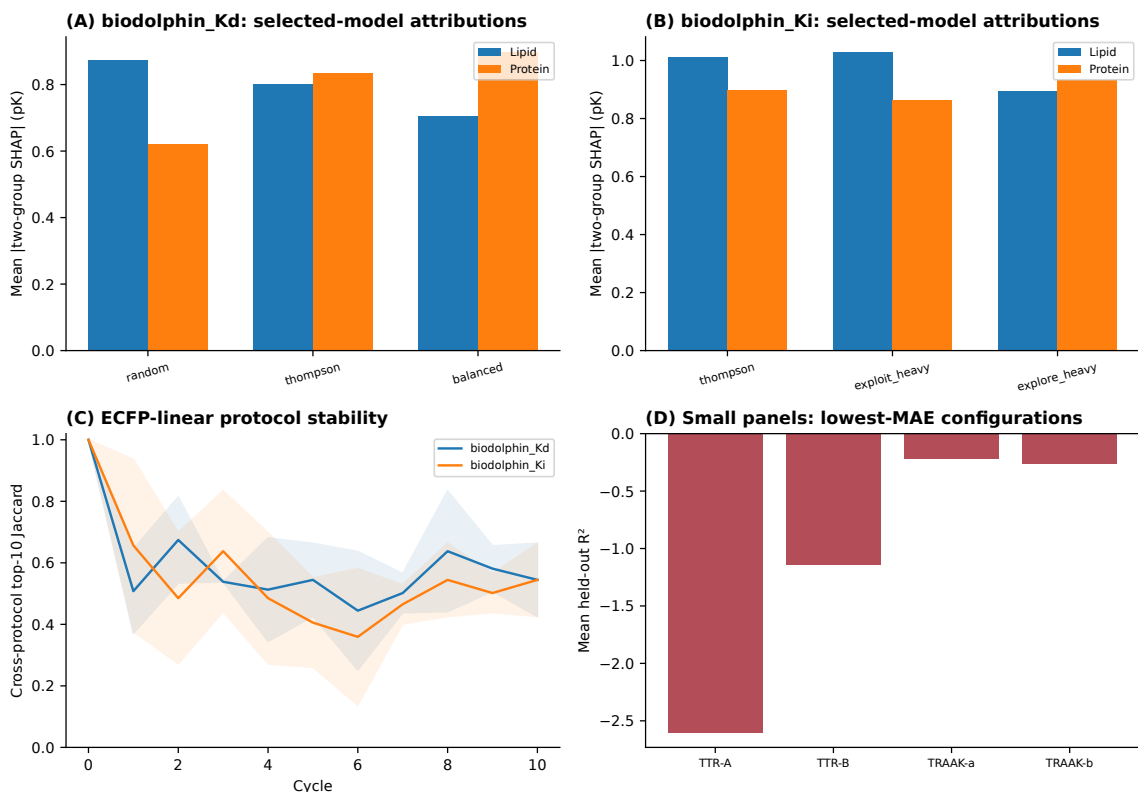


Figure 6: Four-panel interpretability and reliability summary. (A,B) Mean absolute two-player SHAP values for the selected final-model configurations selected from the original grid, averaged across seeds and their explained subsets; representations/kernels differ between selections. (C) Matched ECFP-linear protocol overlap, with seed SD bands. (D) Negative mean R^2 persists on all small panels even for lowest-MAE configurations.

7. Interactive Dashboard and Deployment Scope

The LipidLab portal (Figure 7) exposes all 5,850 benchmark runs through linked dataset, representation, kernel, policy and seed controls. Four-panel curves show held-out MAE, discovery recall, RMSE and interval coverage. A matched representation table reports seed means and SD; audit downloads expose numerical results. Separate views retain 90 explanation reports (54 grouped and 36 temporal), explicitly marking missing SHAP rather than equating a benchmark result with an explanation.

The local Flask service accepts measured CSV pairs, validates sequences, SMILES, affinity units and budgets, and runs retrospective AL using the same supported encoders, kernels and policies. Test rows remain excluded from acquisition; progress, cancellation and result downloads are available. New uploads do not automatically generate SHAP or prospective experimental labels.

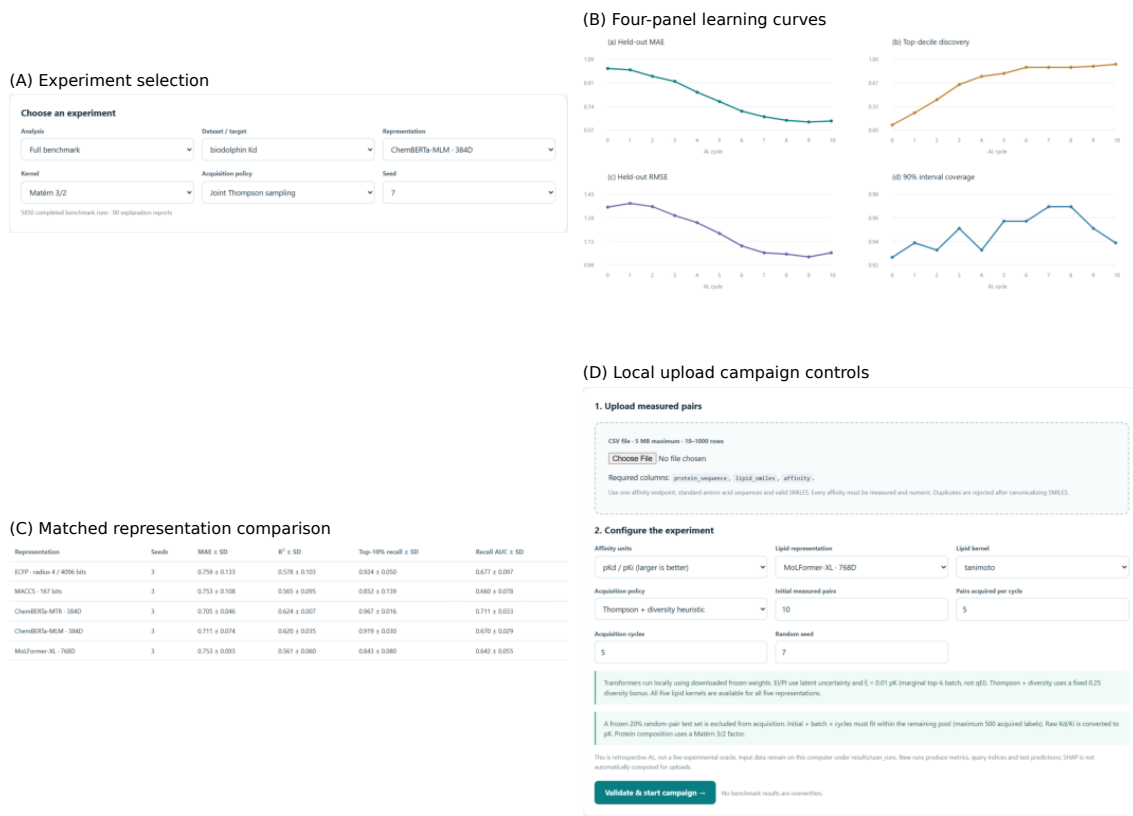


Figure 7: Four screenshots of the actual local dashboard: (A) linked experiment selection, (B) learning-curve panels, (C) matched representation comparison, and (D) upload/campaign controls. The displayed results come from completed published-data experiments; the controls do not imply a public hosted AL service. The screenshot source and hashes accompany the source package.

A read-only static export is prepared for Vercel hosting. It excludes user uploads, pre-trained weights and the Python execution environment. A public deployment has not been verified and no public URL is claimed in this draft. The screenshots show the functioning local application. The current in-memory job queue and local output files are not a durable hosted-job architecture; online AL would require a separately provisioned service with persistent job state, storage, authentication and resource limits. Merely hosting the static page does not make its upload runner executable. An anonymous hosting URL should be added only after deployment and review of data redistribution and conference anonymity requirements.

8. Limitations and Discussion

Generalization and selection. Random-pair splits permit homologous proteins and related chemical structures across train and test. They do not measure cold-target, cold-

family or scaffold generalization. Nested selection, grouped splits and more seeds are needed before selecting a deployment model. The configuration grid is a controlled comparison, not 1,950 independent replications. TTR panels overlap the pooled data. Uncertainty calibration is recorded but not sufficient to establish out-of-distribution reliability.

Assay and dataset scope. Heterogeneous experimental conditions and consensus labels can introduce unmodeled variation. BioDolphin’s structural-resource scale must not be equated to the number of independent measured affinities used here. The broad lipid definition also prevents interpreting every result as membrane-phospholipid recognition. Dedicated larger, homogeneous target-specific affinity panels would strengthen the benchmark.

Explanation scope. Marginal feature replacement may create invalid fingerprint combinations. Protein-conditioned pooled explanations aggregate over different targets, so a recurring bit need not be a universal binding determinant. A representative fragment cannot resolve every collision. Numerical additivity neither establishes causal necessity nor verifies a binding site. No contact maps, docking validation, residue importance or prospective mutagenesis results are claimed. Background-sensitivity and fixed-population temporal controls would help separate model evolution from population drift.

Methodological scope. The most extensive per-bit analysis uses only the linear lipid kernel. Extending ungrouped explanations to non-linear kernels requires additional computation and convergence checks rather than reuse of the affine shortcut. The protein product kernel and optimizer differ from the reference study. Our evidence supports a transparent adaptation and objective trade-off, not a new theory of active learning or a universally optimal policy.

9. Conclusion

In this retrospective protein–lipid study, acquisition enrichment and global predictive accuracy are distinguishable outcomes. Thompson sampling and scheduled UCB can prioritize high-affinity pairs while uncertainty-only querying gives the lowest descriptive prediction error in the expanded grid. Adding frozen transformers does not guarantee improvement. Small target panels remain insufficient for reliable predictive claims. Temporal ECFP attributions and fragment annotations make model behavior inspectable, but their fidelity must not be confused with biological validity. Stronger evidence requires target-aware evaluation, homogeneous measurements, robust explanation controls and prospective testing.

Reproducibility and Supplementary Material

The accompanying source package contains this manuscript, seven four-panel figures and the complete figure atlas. The atlas includes every one of 223 pre-existing result figures in 45 composite figures with four or five source panels each, including legacy-budget plots, full-grid heatmaps, 54-model grouped explanations, and temporal/chemical plots for all 36 campaigns. A CSV manifest maps every panel to its source and SHA-256 hash. Full expanded-grid summaries (1,950 configurations), per-seed outcomes (5,850 campaigns), tied-winner tables and execution audit accompany the paper. The new representation and dashboard

composites are separate from the legacy 223-source atlas. Original checkpoints and measurements are not modified by manuscript generation. Configuration and data hashes identify the computations. The separate atlas is supplementary material, not part of the main-paper page budget, and does not substitute for the evidence summarized above.

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